# From Accounts to Alerts: EDA and Logistic Regression for Early Corporate Risk Prediction

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2025/10/13

# Outline

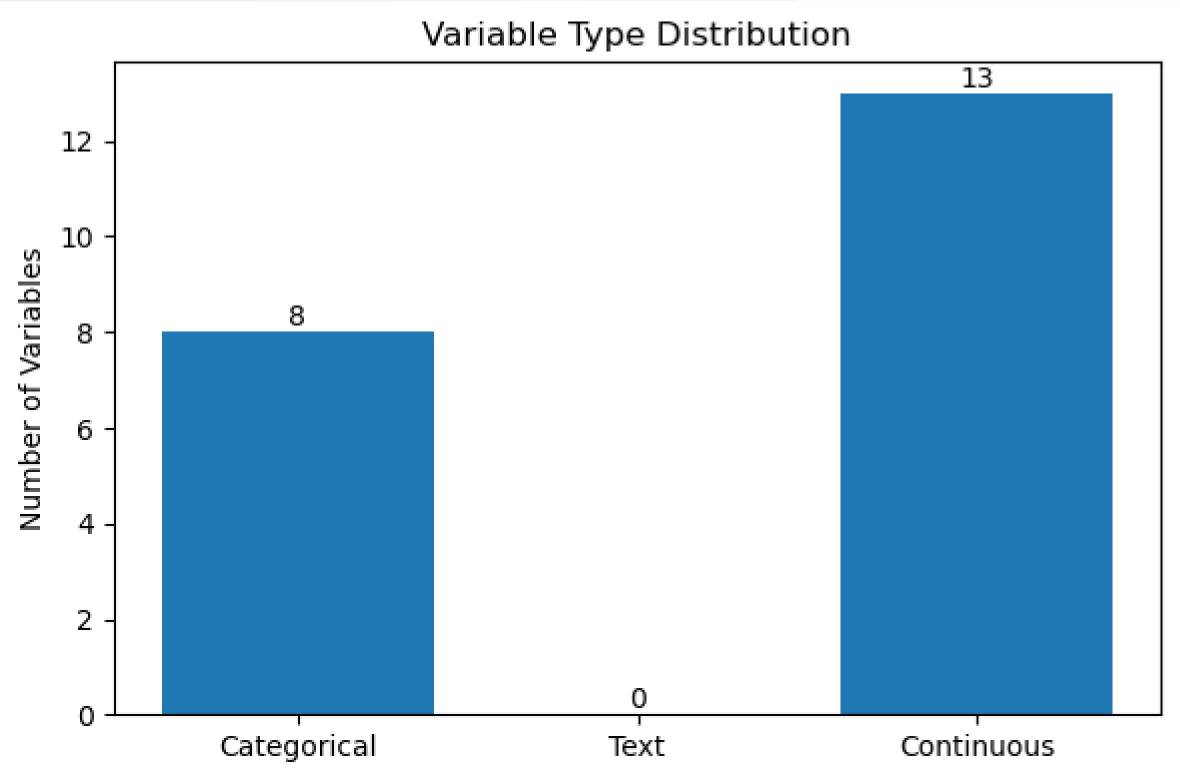
1. Motivation
2. Data
3. Methodology

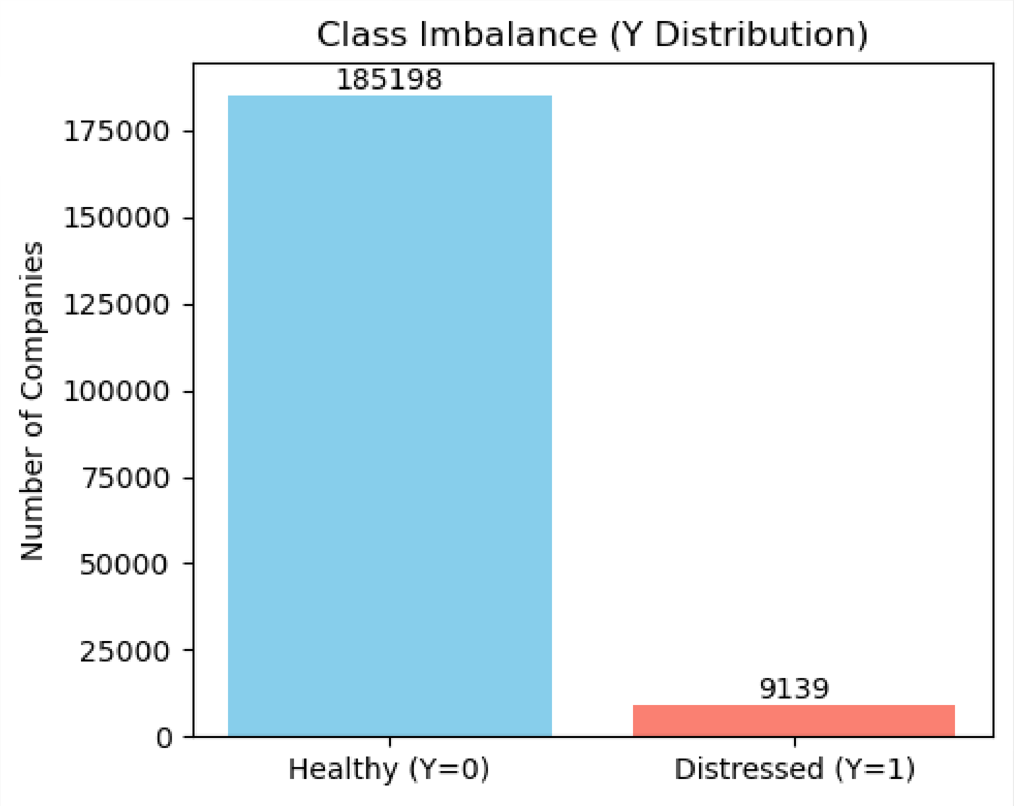
# Motivation

- Early Warning Systems
- Transparency and Interpretability
- Publicly Available Data

# Data

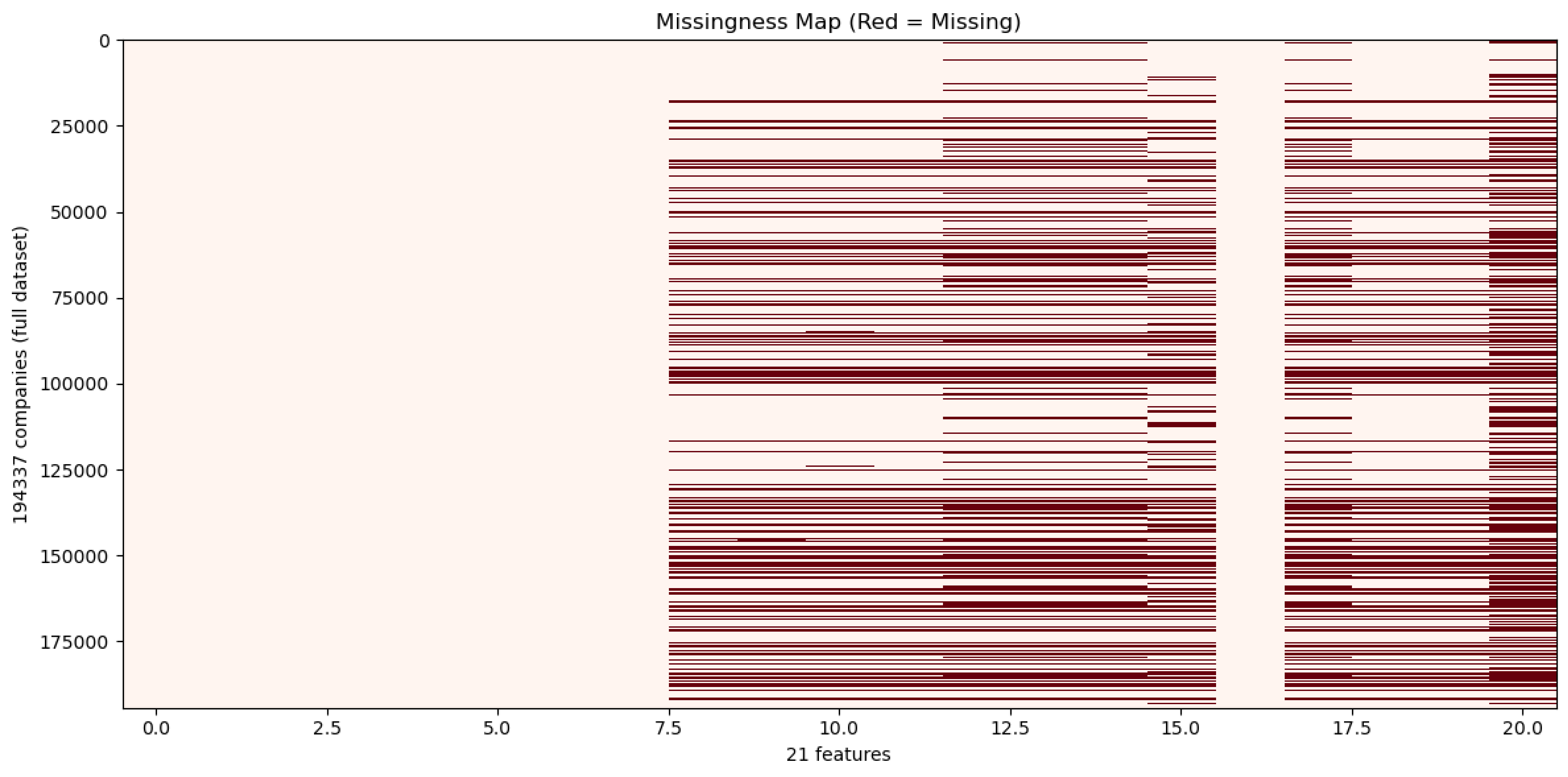
- companies: 23033
- features: 21
- Class imbalance (Y=1 ratio): 0.0470



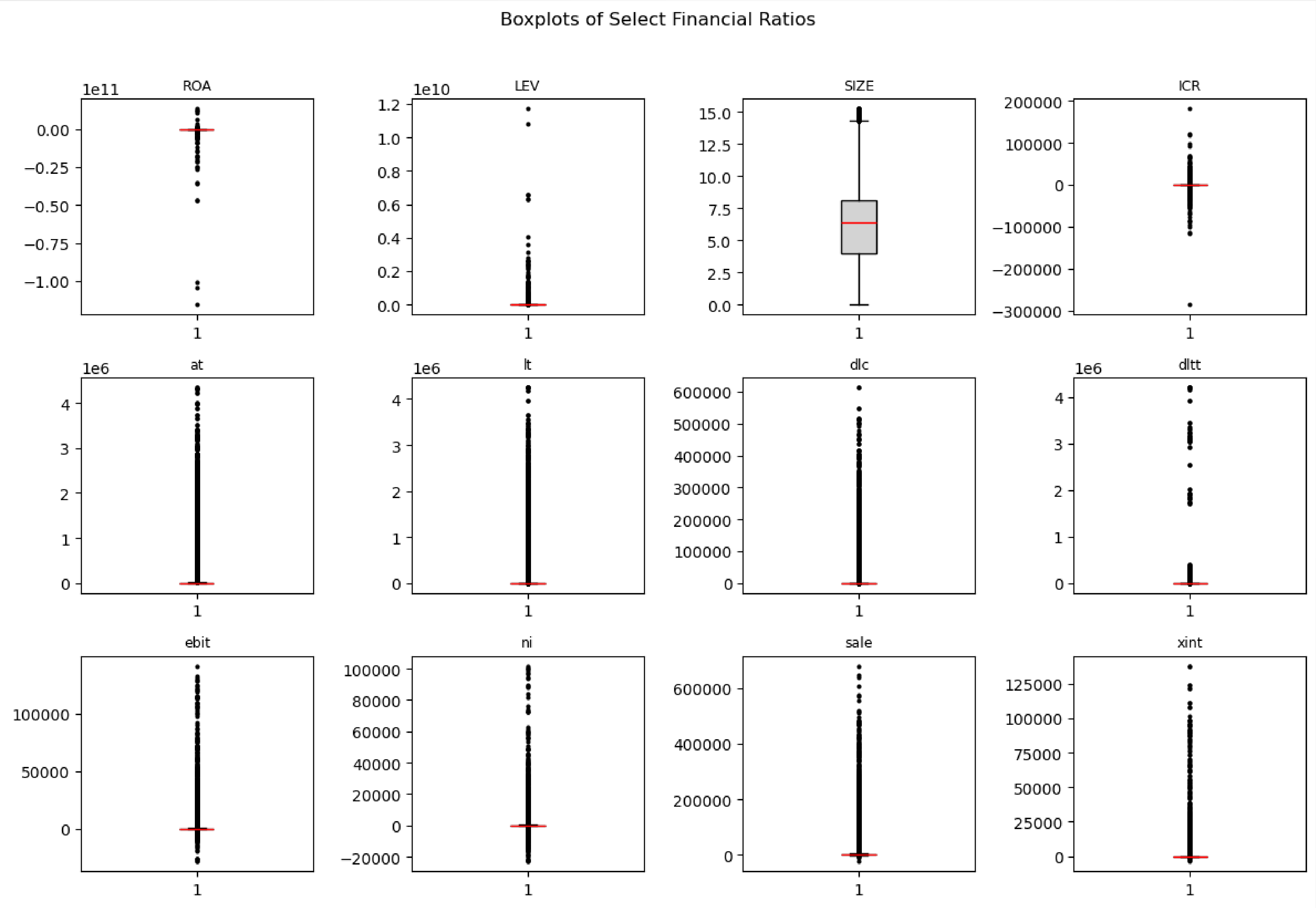
# Data

- Average missing rate across all features: 18.17%



# Data

- heavy-tailed and asymmetric distributions



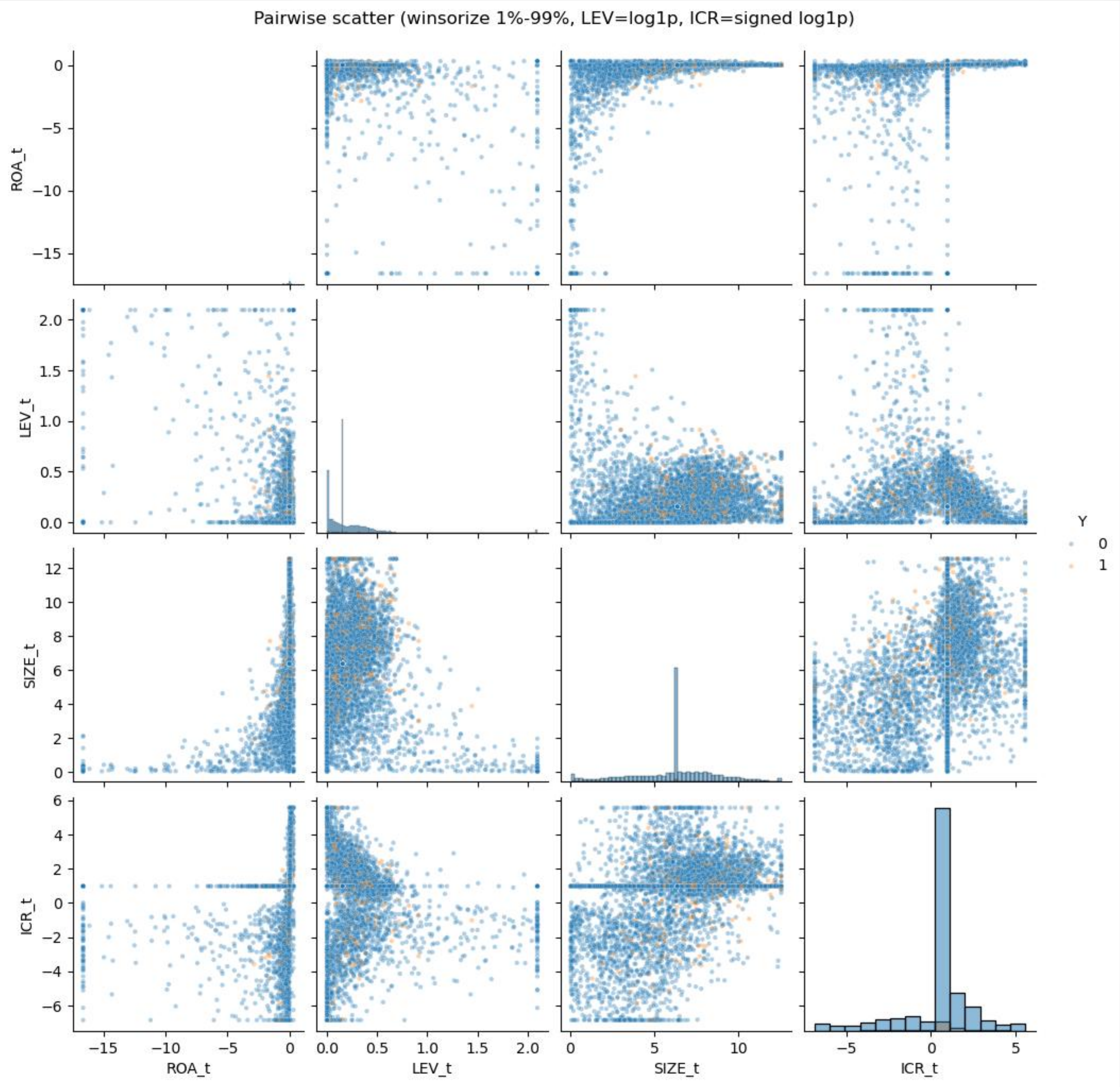
# Data

- Most variables are highly skewed
- Several variables exhibit extreme kurtosis (>10)
- apply winsorization and log-transform before model estimation

[Descriptive Summary of Selected Numerical Features]

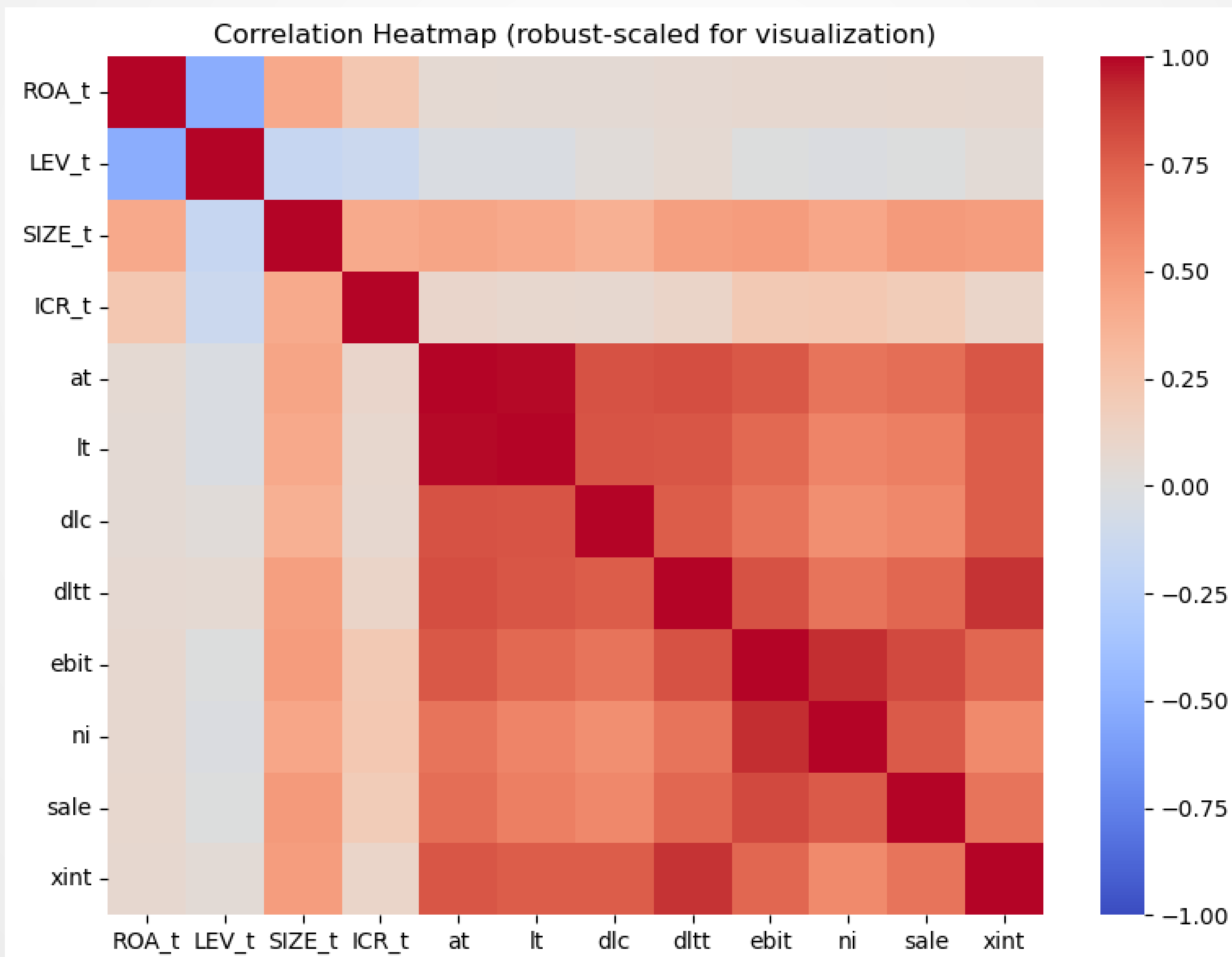
| | count | mean | std | min | 25% | 50% | 75% | max | skewness | kurtosis |
|---|---|---|---|---|---|---|---|---|---|---|
| ROA | 126632.0 | -6869530.06 | 6.125031e+08 | -1.154960e+11 | -0.19 | 0.00 | 0.05 | 1.350200e+10 | -140.97 | 23346.64 |
| LEV | 143774.0 | 1144596.71 | 6.431927e+07 | 0.000000e+00 | 0.03 | 0.17 | 0.41 | 1.172300e+10 | 113.36 | 16453.89 |
| SIZE | 143956.0 | 6.12 | 2.960000e+00 | 0.000000e+00 | 4.02 | 6.37 | 8.14 | 1.529000e+01 | -0.08 | -0.35 |
| ICR | 94503.0 | -54.03 | 2.173370e+03 | -2.855470e+05 | -4.85 | 1.67 | 6.85 | 1.817916e+05 | -19.70 | 4470.59 |
| at | 143956.0 | 20730.17 | 1.526165e+05 | 0.000000e+00 | 54.88 | 582.78 | 3439.39 | 4.349731e+06 | 14.35 | 247.71 |
| lt | 143773.0 | 17758.46 | 1.431946e+05 | -1.000000e-01 | 17.28 | 315.03 | 2244.00 | 4.255074e+06 | 14.88 | 266.24 |
| dlc | 142956.0 | 1433.99 | 1.560927e+04 | -5.600000e+01 | 0.00 | 4.41 | 62.90 | 6.142374e+05 | 19.34 | 449.17 |
| dltt | 143605.0 | 3902.35 | 6.283327e+04 | -2.000000e-02 | 0.04 | 34.15 | 669.61 | 4.216909e+06 | 49.55 | 2719.41 |
| ebit | 126630.0 | 528.02 | 3.263470e+03 | -2.838700e+04 | -4.10 | 6.86 | 143.73 | 1.412900e+05 | 18.27 | 499.06 |
| ni | 126636.0 | 271.61 | 1.971740e+03 | -2.311900e+04 | -9.21 | 1.17 | 68.64 | 1.018320e+05 | 20.72 | 746.89 |
| sale | 126639.0 | 3703.27 | 1.790896e+04 | -2.495468e+04 | 7.12 | 141.59 | 1219.69 | 6.809850e+05 | 13.86 | 283.16 |
| xint | 127418.0 | 211.92 | 2.356540e+03 | -3.775000e+03 | 0.16 | 4.70 | 51.11 | 1.378610e+05 | 31.93 | 1255.55 |

# Data


Pairwise scatter (winsorize 1%-99%, LEV=log1p, ICR=signed log1p)
ROA_t
LEV_t
SIZE_t
ICR_t
Y
0
1

# Data

- Positive correlations among size-related variables (*at*, *lt*, *sale*, *ni*)
- ROA inversely related to LEV (higher leverage, lower profitability)
- No major multicollinearity after data transformation



# Overview

- USA annual financial statements (2019–2024) + SEC Form 25 delisting labels
- Label: Y=1 if a Form 25 becomes effective within 400 days after the financial statement date
- Train: 2019–2023, Test: 2024

# Benchmark Model

- Penalized logistic regression (L1 / Lasso and L2 / Ridge variants)
- Penalization reduces overfitting and selects relevant predictors

# Class Imbalance Handling

- Synthetic Minority Oversampling Technique (SMOTE) applied
- Cost-sensitive weighting adjusts loss for rare default events

# Design of Research

- EDA and Feature Engineering: cleaning, transforming, visualizing ratios
- Model Estimation: penalized logistic regression stratified by sector and time
- Evaluation and Interpretation: performance metrics and SHAP analysis

# Performance Measures

- Discrimination: AUC / ROC, H-measure, and Precision–Recall curves
- Calibration: Brier score and calibration plots
- Threshold-based metrics: accuracy, sensitivity, specificity, F1-score

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